

```
print(rev('Hello, world!'))
print(rev(''))
print(rev('X'))
```

And here is the equivalent JavaScript code in *reverseString.html*:

JavaScript

```
<script type="text/javascript">
function rev(theString) {
  ❶ if (theString.length === 0 || theString.length === 1) {
    // BASE CASE
    return theString;
  } else {
    // RECURSIVE CASE
    ❷ var head = theString[0];
    ❸ var tail = theString.substring(1, theString.length);
    ❹ return rev(tail) + head;
  }
}

document.write(rev("abcdef") + "<br />");
document.write(rev("Hello, world!") + "<br />");
document.write(rev("") + "<br />");
document.write(rev("X") + "<br />");
</script>
```

Here is the output of these programs:

```
fedcba
!dlrow ,olleH

X
```

Our recursive function `rev()` returns the string that is the reverse of the argument, `theString`. Let's consider the simplest strings to reverse: the empty string and a single-character string would “reverse” to themselves. These are the two base cases with which we'll start (though we combine them with an `or` or `||` Boolean operator ❶). For the recursive case, we form `head` from the first character in `theString` ❷, and `tail` from every character after the first ❸. The recursive case then returns the reverse of `tail` followed by the `head` character ❹.